PLSC 40601

Week 2: Sample-splitting, bagging, (honesty).

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Housekeeping

- ?

Sample splitting

What is our goal in fitting a model?

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- What should the expectation be taken over? Can/should we hold the data we used for fitting the model fixed?

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- We may be interested in $\mathrm{Err}_{\mathcal{T}}$, in practice most estimating methods will give us estimates of Err.

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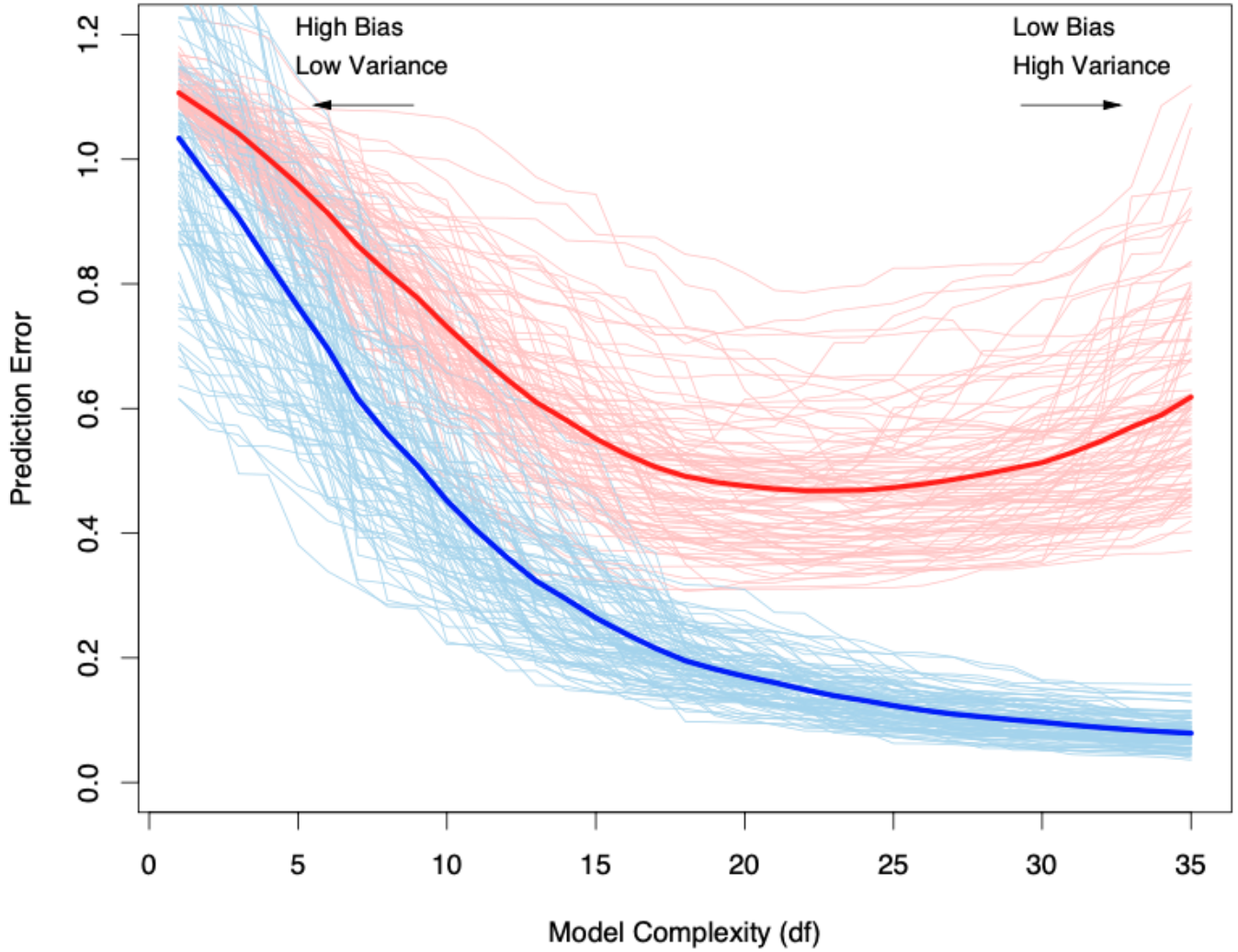

Hastie et al. (2009)

Blue is in-sample error, red is out-of-sample error.

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- And, once we have selected a version of a model, we may want to **assess** how a selected model performs.

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- We can't measure expected test error directly.

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- Why do we need to estimate the prediction error of the selected model *again*? Still a version of winner's curse.

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K-fold cross validation.

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Version 4	Training	Validation	Training	Training	Training
Version 5	Validation	Training	Training	Training	Training

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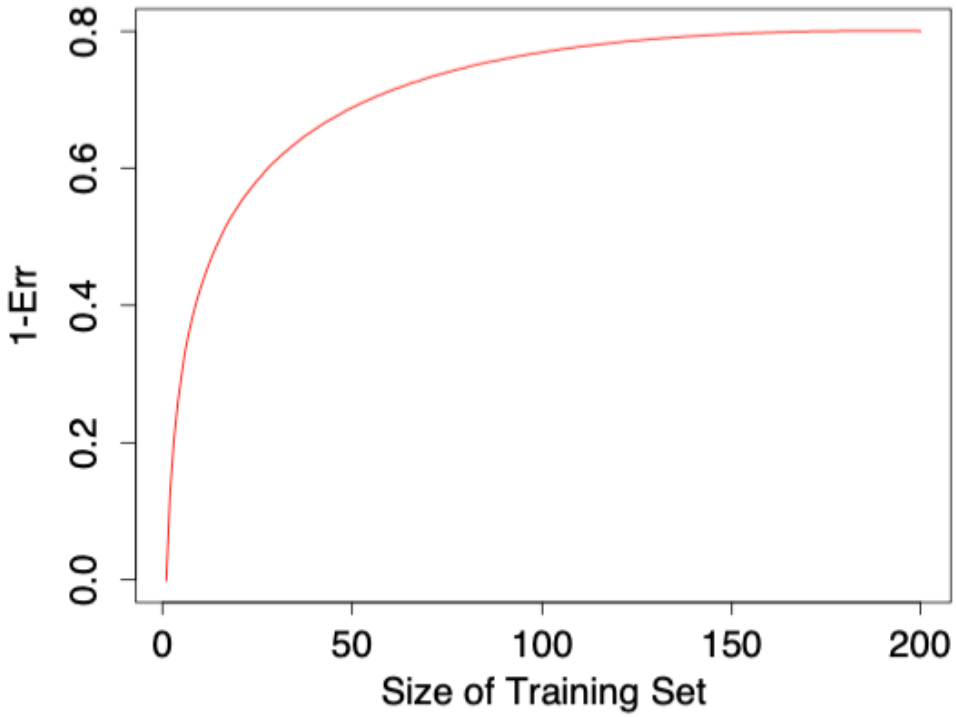

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- Rule of thumb is often 5 or 10.

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- While we do not, we can *suppose* that the empirical CDF produced by the data that we observe is identical to the population CDF.

- We can then just resample with replacement from our observed data, and see how much our estimates vary across resamples.

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- To get an estimate of the standard error of the estimate, calculate the standard deviation over these many bootstrap estimates.

Bootstrapping

- How can we translate this method to estimate test error?

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$$\widehat{\mathrm{Err}}_{Boot} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{|C^{-i}|} \sum_{b \in C^{-i}} L(y_i, \hat{f}^b(x_i))$$

C^{-i} is the set of bootstrap samples that do not contain i, $|C^{-i}|$ is the size of this set.

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  where $\overline{\mathrm{err}}$ is the training error.
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- Some alternatives in Hastie et al. (2009).

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- Bootstrap aggregating.

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- The bagging estimate is

$$\hat{f}_{bag}(x) = \frac{1}{B} \sum_{i=1}^{N} \hat{f}^b(x)$$

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- If we're using a classifier, each model can get a "vote" for each x, and the class with the most votes wins.

- Or we can use averages of classifiers to produce probabilities, rather than just class predictions.

Bagging

- How many bootstrap replicates?

Bagging

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- 25? 50?

Bagging

- How many bootstrap replicates?
- 25? 50? See how your results change with more replicates.

Honesty.

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- Returning to (causal) inference…we might like to use these methods to get valid inference, potentially on causal targets.

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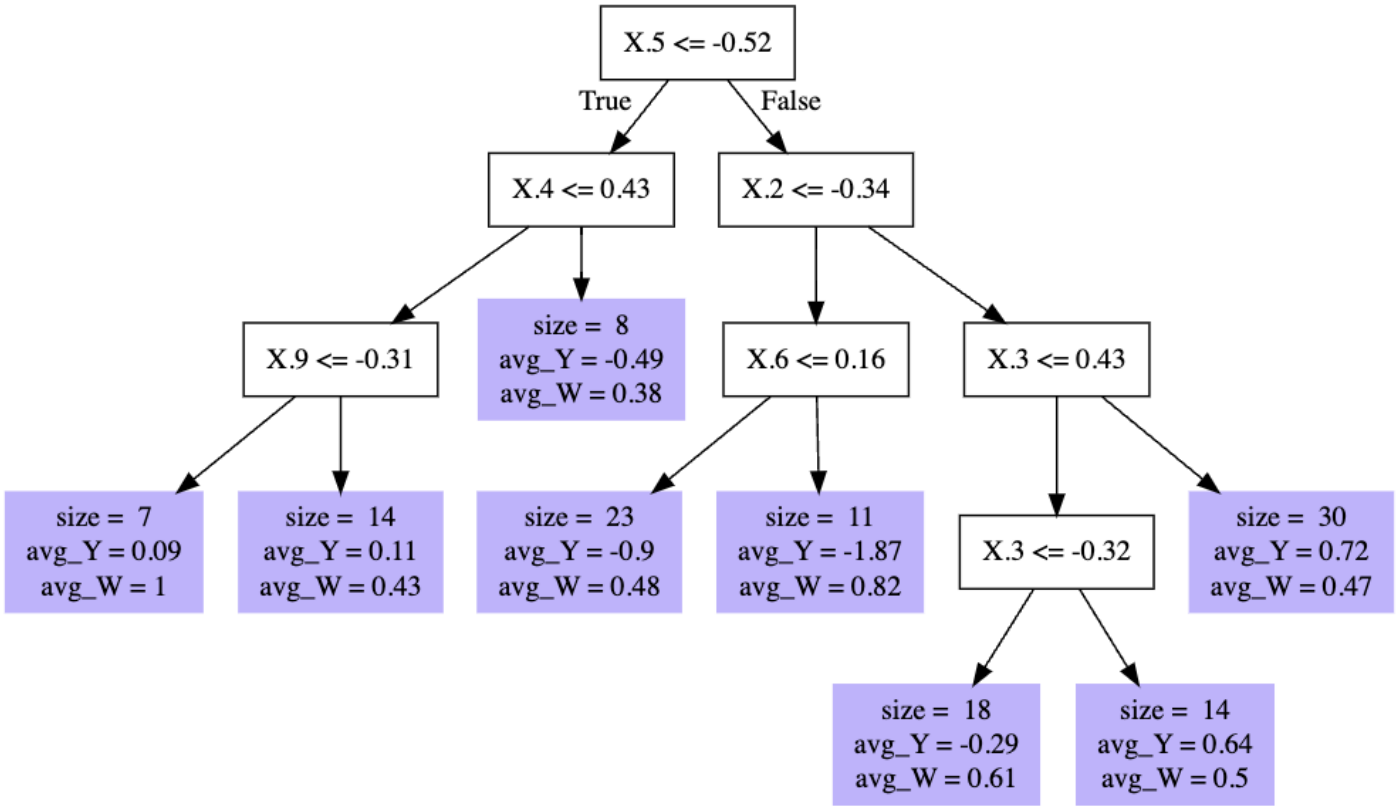
- This procedure reduces bias relative to those proposed by Breiman (2001).

An honest tree algorithm

```
> library(grf)
> set.seed(60637)
> n <- 500
> p <- 10
> X <- matrix(rnorm(n * p), n, p)
> W <- rbinom(n, 1, 0.5)
> Y <- pmax(X[, 1], 0) * W + X[, 2] +
+   pmin(X[, 3], 0) + rnorm(n)
> c.forest <- causal_forest(X, Y, W)
```

An honest tree



References I

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Hastie, T., Tibshirani, R., and Friedman, J. H. (2009). *The elements of statistical learning: data mining, inference, and prediction*, volume 2. Springer.